Chapter II.5, Week 7

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5.9. Let S be a graded ring, generated by S_1 as an S_0-algebra, M a graded S-module, and $X = \operatorname{Proj} S$.

(a) Show there is a natural homomorphism $\alpha : M \to \Gamma_*(\widetilde{M})$.

(b) Assume now that $S_0 = A$ is a finitely generated k-algebra for some field k, that S_1 is a finitely generated A-module, and that M is a finitely generated S-module. Show that the map α is an isomorphism in all large enough degrees, i.e., there is a $d_0 \in \mathbb{Z}$ such that for all $d \geqslant d_0$, $\alpha_d : M_d \to \Gamma(X, \widetilde{M}(d))$ is an isomorphism. [*Hint*: Use the methods of the proof of (5.19).]

(c) With the same hypothesis, we define an equivalence relation $\approx$ on graded S-modules by saying $M \approx M'$ if there is an integer d such that $M_{\geqslant d} \cong M'_{\geqslant d}$. Here, $M_{\geqslant d} = \bigoplus_{n \geqslant d} M_n$. We say that a graded S-module is *quasi-finitely generated* if it is equivalent to a finitely generated module. Now show that the functors $\widetilde{\cdot}$ and Γ_* induce an equivalence of categories between the category of quasi-finitely generated graded S-modules modulo the equivalence relation $\approx$, and the category of coherent $\mathcal{O}_X$-modules.

Proof. (a) Fix generators $\{x_i\}_{i \in I} \subset S_1$ of S as an S_0-algebra. This gives us an affine open cover $\{D_+(x_i) \cong \operatorname{Spec} S_{(x_i)}\}_{i \in I}$ of $\operatorname{Proj} S$ such that

$$\begin{aligned} \widetilde{M}|_{D_+(x_i)}(n) &\cong \widetilde{M(n)}|_{D_+(x_i)} \\ &\cong (M(n)_{(x_i)})^{\sim} \\ &\cong ((M_{x_i})_n)^{\sim} \end{aligned}$$

since restriction commute with tensor products, and $x_i \in S_1$ implies

$$(M \otimes_S N)_{(x_i)} = M_{(x_i)} \otimes_{S_{(x_i)}} N_{(x_i)}.$$

Then

$$\begin{aligned} \Gamma_*(\widetilde{M}|_{D_+(x_i)}) &\cong \bigoplus_{n \in \mathbb{Z}} \Gamma(D_+(x_i), \widetilde{M}|_{D_+(x_i)}(n)) \\ &\cong \bigoplus_{n \in \mathbb{Z}} \Gamma(\operatorname{Spec} S_{(x_i)}, ((M_{x_i})_n)^{\sim}) \\ &\cong \bigoplus_{n \in \mathbb{Z}} (M_{x_i})_n \\ &\cong M_{x_i} \end{aligned}$$

supplies us with a natural map $\alpha_i : M \hookrightarrow M_{x_i} \xrightarrow{\sim} \Gamma_*(\widetilde{M}|_{D_+(x_i)})$ for each i. Using a similar argument to the above, we define maps $\alpha_{ij} : M \hookrightarrow M_{x_i x_j} \xrightarrow{\sim} \Gamma_+(\widetilde{M}|_{D_+(x_i x_j)})$ and show that $\rho_{D_+(x_i x_j)} \circ \alpha_i = \alpha_{ij} = \rho_{D_+(x_i x_j)} \circ \alpha_j$.

Fixing $m \in M$, the $\alpha_i(m)_n \in \Gamma(D_+(x_i), \widetilde{M}|_{D_+(x_i}(n))$ glue together to a section $\alpha(m)_n \in \Gamma(X, \widetilde{M})$. Only finitely many $\alpha_i(m)_n$ are nonzero for each i, so only finitely many $\alpha(m)_n$ are nonzero, whence $\alpha(m) = \sum_n \alpha(m)_n \in \Gamma_*(\widetilde{M})$ is well-defined. We take $\alpha : m \mapsto \alpha(m)$ to be our natural map.

(b) By (I, 7.4), we have a finite filtration

$$0 = M^0 \subset M^1 \subset \cdots \subset M^r = M$$

of M by graded submodules, where for each i, $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n_i)$ for some homogeneous prime ideals $\mathfrak{p}_i \trianglelefteqslant S$ and some integer n_i. Combining this with exactness of $\widetilde{\cdot}$, left exactness of $\Gamma(X, \cdot)$, and exactness of taking degree gives us the following diagram of S_0-modules with exact rows, for any $d \in \mathbb{Z}$.

$$\begin{tikzcd}
0 \arrow[r] & M_d^{i-1} \arrow[r] \arrow[d] & M_d^i \arrow[r] \arrow[d] & M^i/M_d^{i-1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \Gamma(X, \widetilde{M}^{i-1}(d)) \arrow[r] & \Gamma(X, \widetilde{M}^i(d)) \arrow[r] & \Gamma(X, \widetilde{M^i/M^{i-1}}(d)) &
\end{tikzcd}$$

If we have that the left and right vertical maps are isomorphisms, the five lemma implies the middle map is an isomorphism.

The result clearly holds for $M^0 = 0$, so it suffices to show the result for for the $M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n)$. Our map at degree d is an isomorphism iff it is a bijection, allowing us to disregard our original S and reduce to the case of $M = S$ a graded integral domain, finitely generated by S_1 as an S_0-algebra, and $S_0 = A$ a finitely generated integral domain over k.

Take such S, with $x_0, \ldots, x_r \in S_1$ generators as an A-module. Define $T_{\geqslant n} := \bigoplus_{d \geqslant n} T_n$ for any graded ring T, and denote $S' := \Gamma_*(\mathcal{O}_X)_{\geqslant 0}$. We will show that

$$\alpha_{\geqslant m} : S_{\geqslant m} \to S'_{\geqslant m}$$

is an isomorphism for $m \gg 0$. S is an integral domain, so the localization maps $S \to S_f$ are injective for all $f \neq 0$. In particular, each of the $S \to S_{x_i}$, $S_{x_i} \to S_{x_i x_j}$ are injective, as is $S \to S_{x_0 \cdots x_r}$. Thus, $\alpha_{d \geqslant 0} : S \to S'$ corresponds to the inclusion $S \subset S' \subset \cap_i S_{x_i}$. It suffices to show $S'_{\geqslant m} \subset S_{\geqslant m}$ for $m \gg 0$.

By the proof of 5.19, we have that S' is a finitely generated S-module. Take such a set of homogeneous generators $y_0, \ldots, y_l$ of degree $d_j \geqslant 0$. Since $y_j \in S_{x_i}$ for all i, we can find a single integer n such that $x_i^n y_j \in S$ for all i. The x_i generate S_1, so the monomials in the x_i of degree m generate S_m for any m. Then we can take larger n such that $f y_j \in S$ for all $f \in S_n$. Since y_j is of positive degree, $f y_j \in S_{\geqslant n} := \bigoplus_{d \geqslant n} S_d$ whenever $f \in S_{\geqslant n}$. Take one n that works for all y_j, so that $S_{\geqslant n} \subset S_{\geqslant n}[y_0, \ldots, y_l] \subset S_{\geqslant n}$. Then take $m = n + \max_j d_j$, so that

$$S'_{\geqslant m} \subset S_{\geqslant n}[y_0, \ldots, y_l] = S_{\geqslant n} \subset S_{\geqslant m}.$$

We conclude that $\alpha_{\geqslant m} : S_{\geqslant m} \to S'_{\geqslant m}$ is an isomorphism for $m \gg 0$, as desired.

(c) If $M_d = 0$ for $d \gg 0$, then any $m \in \Gamma(X, \widetilde{M})$ restricts to $m_i \in M_{x_i}$ with $m = \frac{x_i^d m}{x_i^d} = 0$, whence $\widetilde{M} = 0$. Note that $\ker(\phi_d) = (\ker \phi)_d$ and $\operatorname{cok}(\phi_d) = (\ker \phi)_d$, so the following sequence is unambiguous and exact.

$$\ker \phi_{\geqslant d} \longrightarrow M_{\geqslant d} \xrightarrow{\phi_{\geqslant d}} M'_{\geqslant d} \longrightarrow \operatorname{cok} \phi_{\geqslant d}$$

Then if $\phi_{\geqslant d} : M_{\geqslant d} \to M'_{\geqslant d}$ is an isomorphism for $d \gg 0$, exactness of $\widetilde{\cdot}$ gives us

$$0 = \widetilde{\ker \phi_{\geqslant d}} \longrightarrow \widetilde{M}_{\geqslant d} \xrightarrow{\phi_{\geqslant d}} \widetilde{M'}_{\geqslant d} \longrightarrow \widetilde{\operatorname{cok} \phi_{\geqslant d}} = 0$$

is exact, whence $\widetilde{M} \cong \widetilde{M'}$. Thus, $\widetilde{\cdot}$ induces a map from $\approx$-classes to quasi-coherent $\mathcal{O}_X$-modules. Under the assumption that S is noetherian, $\widetilde{\cdot}$ induces a map from $\approx$-classes of quasi-finitely generated S-modules to coherent $\mathcal{O}_X$-modules.

We can use 5.19 to show that $\Gamma_*(\cdot)$ induces a map from coherent $\mathcal{O}_X$-modules to $\approx$-classes of quasi-finitely generated S-modules as well. By 5.15, we have a natural isomorphism $\beta : \Gamma_*(\cdot)^\sim \to \mathrm{id}$, and by (b) we have a natural isomorphism of $\approx$-classes $\alpha : \mathrm{id} \to \Gamma_*(\widetilde{\cdot})$. We conclude that $\Gamma_*(\cdot)$ and $\widetilde{\cdot}$ induce an equivalence of categories, as desired. $\square$

5.10. Let A be a ring, $S = A[x_0, \ldots, x_r]$, and $X = \operatorname{Proj} S$. We have seen that a homogeneous ideal I in S defines a closed subscheme of X (Ex. 3.12), and that convrsely every closed subscheme of X arises in this way (5.16).

(a) For any homogeneous ideal $I \subset S$, we define the saturation $\overline{I}$ of I to be

$$\overline{I} := \{s \in S \mid \forall 0 \leqslant i \leqslant r \ (\exists n > 0 \ (x_i^n s \in I))\}$$

We say that I is *saturated* if $I = \overline{I}$. Show that $\overline{I}$ is a homogeneous ideal of S.

(b) Two homogeneous ideals I_1 and I_2 of S define the same closed subscheme of X iff they have the same saturation.

(c) If Y is any closed subscheme of X, then the ideal $\Gamma_*(\mathscr{I}_Y)$ is saturated. Hence, it is the largest homogeneous ideal defining the subscheme Y.

(d) There is a 1-1 correspondence between saturated ideals of S and closed subschemes of X.

Proof. (a) If $s \in \overline{I}$, then we can write $s = \sum_d s_d$ as the sum of its homogeneous parts, so that $x_i^n s = \sum_d x_i^n s_d \in I$ and I homogeneous implies $x_i^n s_d \in I$ for each d. We conclude that $s_d \in \overline{I}$ for all d, whence $\overline{I}$ is homogeneous.

(b) Fix homogeneous ideals $I_1, I_2 \trianglelefteq S$. We have an affine cover $D_+(x_i) \cong \operatorname{Spec} S_{(x_i)}$ of X, so it suffices to show that

$$\frac{S_{(x_i)}}{(I_1)_{(x_i)}} \cong \widetilde{S/I_1}|_{D_+(x_i)} \cong \widetilde{S/I_2}|_{D_+(x_i)} \cong \frac{S_{(x_i)}}{(I_2)_{(x_i)}}$$

for all $0 \leqslant i \leqslant r$ iff $\overline{I}_1 = \overline{I}_2$. This occurs iff the kernels of the respective projection maps are the same, i.e. $(I_1)_{(x_i)} = (I_2)_{(x_i)}$. Clearly this holds if $(I_1)_{x_i} = (I_2)_{x_i}$, so suppose conversely that the equality holds only in degree 0. Then x_i is a unit, so

$$\begin{aligned} f \in ((I_1)_{x_i})_d &\iff f/x_i^d \in (I_1)_{x_i} = (I_2)_{x_i} \\ &\iff f \in ((I_2)_{x_i})_d \end{aligned}$$

gives us that $(I_1)_{x_i} = (I_2)_{x_i}$.

We have that

$$S \cap (I_1)_{x_i} = \{s \in S : \exists n > 0 \ (x_i^n s \in I_1)\},$$

so

$$\overline{I}_1 = S \cap \bigcap_{i=0}^r (I_1)_{x_i} = S \cap \bigcap_{i=0}^r (I_2)_{x_i} = \overline{I}_2$$

whenever the equality of localizations holds. Conversely, suppose $(I_1)_{x_i} \neq (I_2)_{x_i}$ so, without loss of generality, there exists $s \in (I_1)_{x_i} \setminus (I_2)_{x_i}$. Then we can take $n > 0$ such that $x_i^n s \in I_1 \subset \overline{I}_1$, but $x_i^m(x_i^n s) \notin I_2$ for all $m > 0$ implies $x_i^n s \in \overline{I}_1 \setminus \overline{I}_2$, whence $\overline{I}_1 \neq \overline{I}_2$.

(c) We have that $\mathscr{I}_Y = \ker(i^\sharp : \mathcal{O}_Y \to i_*\mathcal{O}_X)$ is a subsheaf of $\mathcal{O}_Y$, so

$$J_i := \Gamma(D_+(x_i), \mathscr{I}_Y) \subset \Gamma(D_+(x_i), \mathcal{O}_Y) = S_{(x_i)}$$

implies

$$J_i(n) \cong \Gamma(D_+(x_i), \mathscr{I}_Y(n)) \subset \Gamma(D_+(x_i), \mathcal{O}_Y(n)) = S(n)_{(x_i)}$$

for all $n \in \mathbb{Z}$, whence $\Gamma_*(\mathscr{I}_Y) \subset \Gamma_*(\mathcal{O}_X) \cong S$ by 5.15.

Now, fix $s \in \overline{J}_m$, so that $x_i^r s \in J$ for $r >> 0$. Write

$$x_i^m J_i = J_i(m) \subset S(m)_{(x_i)} = x_i^m S_{(x_i)}.$$

We have $s_i := s|_{D_+(x_i)} \in x_i^m S(m)_{(x_i)}$, and $x_i^r s_i \in (\Gamma_*(\mathscr{I}_Y)|_{D_+(x_i)})_{m+r} = J_{m+r}$. Then $x_i^r s_i = x_i^{m+r} j_i$ for $j_i \in J_i$, so $x_i \in J_i^\times$ gives us

$$s_i = x_i^r j_i \in J_m.$$

Then s is a gluing of sections in $s_i \in \Gamma(D_+(x_i), \mathscr{I}_Y(m))$, so $s \in \Gamma(X, \mathscr{I}_Y(m)) \subset \Gamma_*(\mathscr{I}_Y)$, as desired.

(d) To any saturated ideal $I \trianglelefteq S$, we associated the closed subscheme $Y_I := \operatorname{Proj} S/I$. By (b), this map $I \mapsto Y_I$ is injective. By (c), it is surjective, with $\Gamma_*(\mathscr{I}_Y) \mapsto \operatorname{Proj} S/\Gamma_*(\mathscr{I}_Y) \cong Y$ for any closed subscheme Y. This gives us our bijection.

$\square$